

Unit 8, Lesson 6: Solving Multi-Step Equations

Benchmark

MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

Learning Target

☐ I can solve multi-step linear equations with variables on both sides.

8.6.1 Warm-Up: Math Argument

Two friends, Dave and Michelle, play Candy Crave often. Their records are shown below.

Dave’s Scores and Statistics		Michelle’s Scores and Statistics	
Games Played	529	Games Played	457
High Score	8,931	High Score	6,751
Average Score	5,126	Average Score	5,817
Latest Score	7,915	Latest Score	5,284
Longest Game	8 min 3 sec	Longest Game	5 min 9 sec
Total Play Time	12 hr 9 min	Total Play Time	9 hr 13 min

1. Use information from the table to argue why Dave is the better player.
2. Use the same information to argue that Michelle is the better player instead.

8.6.2 Collaboration: Finding the Error

1. Clare was solving an equation, but when she checked her solution, she noticed it was incorrect. She knows she made a mistake but cannot find it.

Clare’s work is shown.

Clare’s Work	Corrected Solution
$12(5 + 2y) = 4y - (5 - 9y)$ $60 + 24y = 4y - 5 - 9y$ $60 + 24y = -5y - 5$ $24y = -5y - 55$ $29y = -55$ $y = \frac{-55}{29}$	

- a. Discuss Clare’s work with your partner to find her mistake. Then, circle where she made the mistake.
- b. Find the correct solution to the equation using the space provided above.
- c. Verify your solution is correct using substitution.

8.6.3 Activity: Solution Card Sort

You and your partner will receive 12 cards for this activity.

- 1. Shuffle the cards and place them face-down in a pile.
- 2. Record your partner’s name. _____
- 3. Each player draws a card from the pile and solves the equation in the corresponding box on the recording sheet below. Be sure to record the work in the correct box by matching the letter on the card to the letter in the box.

A	B
C	D
E	F

G	H
I	J
K	L

- 4. When all equations have been solved, sort the cards into groups based on their number of solutions. Write the letters that correspond to the cards in each group in the table below.

One Solution	No Solution	Infinitely Many Solutions

8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help
someone.

I can solve multi-step linear equations.	  
I can solve equations with variables on both sides.	  
I can relate solutions to equations to real-world contexts they model.	  
I can recognize equations with one, none, or infinitely many solutions.	  